

1.5
28f) $\exists x \forall y (y \neq 0 \rightarrow xy = 1)$ Domain: all real numbers

"There is a real number x such that for every nonzero y , $xy = 1$ "

aka $x = \frac{1}{y}$ for all non zero y

This is false because $\frac{1}{y}$ is different for every y .

1.5
31) a) $\forall x \exists y \forall z T(x, y, z)$

$$\neg \forall x \exists y \forall z T(x, y, z)$$

$$\exists x \neg \exists y \forall z T(x, y, z)$$

$$\exists x \forall y \neg \forall z T(x, y, z)$$

$$\boxed{\exists x \forall y \exists z \neg T(x, y, z)}$$

b) $\forall x \exists y (P(x, y) \vee \forall x \exists y Q(x, y))$

$$\neg (\forall x \exists y (P(x, y) \vee \forall x \exists y Q(x, y)))$$

$$\neg \forall x \exists y (P(x, y) \vee \forall x \exists y Q(x, y))$$

$$\exists x \neg \exists y (P(x, y) \vee \forall x \exists y Q(x, y))$$

$$\boxed{\exists x \forall y \neg (P(x, y) \vee \forall x \exists y Q(x, y))}$$

c) $\forall x \exists y (P(x, y) \wedge \exists z R(x, y, z))$

$$\neg \forall x \exists y (P(x, y) \wedge \exists z R(x, y, z))$$

$$\exists x \neg \exists y (P(x, y) \wedge \exists z R(x, y, z))$$

$$\exists x \forall y \neg (P(x, y) \wedge \exists z R(x, y, z))$$

$$\exists x \forall y (\neg P(x, y) \vee \neg \exists z R(x, y, z))$$

$$\boxed{\exists x \forall y (\neg P(x, y) \vee \forall z \neg R(x, y, z))}$$

d) $\forall x \exists y (P(x, y) \rightarrow Q(x, y))$

$$\neg \forall x \exists y (P(x, y) \rightarrow Q(x, y))$$

$$\neg \forall x \exists y (\neg P(x, y) \vee Q(x, y))$$

$$\exists x \neg \exists y (\neg P(x, y) \vee Q(x, y))$$

$$\exists x \forall y \neg (\neg P(x, y) \vee Q(x, y))$$

$$\boxed{\exists x \forall y (P(x, y) \wedge \neg Q(x, y))}$$

1.5
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a) $\neg \forall x \forall y P(x, y)$
 $\exists x \neg \forall y P(x, y)$
 $\boxed{\exists x \exists y \neg P(x, y)}$

b) $\neg \forall x \exists y P(x, y)$
 $\exists x \neg \exists y P(x, y)$
 $\boxed{\exists x \forall y \neg P(x, y)}$

c) $\neg \forall y \forall x (P(x, y) \vee Q(x, y))$
 $\exists y \neg \forall x (P(x, y) \vee Q(x, y))$
 $\exists y \exists x \neg (P(x, y) \vee Q(x, y))$
 $\boxed{\exists y \exists x (\neg P(x, y) \wedge \neg Q(x, y))}$

d) $\neg (\exists x \exists y \neg P(x, y) \wedge \forall x \forall y Q(x, y))$
 $\neg (\exists x \exists y \neg P(x, y) \wedge \forall x \forall y Q(x, y))$
 $\neg \exists x \exists y \neg P(x, y) \vee \neg \forall x \forall y Q(x, y)$
 $\forall x \neg \exists y \neg P(x, y) \vee \exists x \neg \forall y Q(x, y)$
 $\forall x \forall y P(x, y) \vee \exists x \exists y \neg Q(x, y)$
 $\boxed{\forall x \forall y P(x, y) \vee \exists x \exists y \neg Q(x, y)}$

e) $\neg \forall x (\exists y \forall z P(x, y, z) \wedge \exists x \forall y P(x, y, z))$
 $\exists x \neg (\exists y \forall z P(x, y, z) \wedge \exists x \forall y P(x, y, z))$
 $\exists x (\neg \exists y \forall z P(x, y, z) \vee \neg \exists x \forall y P(x, y, z))$
 $\exists x (\forall y \neg \forall z P(x, y, z) \vee \forall x \neg \forall y P(x, y, z))$
 $\boxed{\exists x (\forall y \exists z \neg P(x, y, z) \vee \forall x \exists y \neg P(x, y, z))}$

1.5
39 Domain: all integers

a) $\forall x \forall y (x^2 = y^2 \rightarrow x = y)$

counterexample: $x=2, y=-2$; $2^2 = (-2)^2 = 4$, but $2 \neq -2$.

b) $\forall x \exists y (y^2 = x)$

counterexample: $x=2$, but there is no integer y such that $y^2=2$.
 ($y=\sqrt{2}$ is not an integer)

c) $\forall x \forall y (xy \geq x)$

counterexample: $x=-1, y=2$; $(-1)(2) = -2 \not\geq -1$

(1.5)
40 Domain: all integers

a) $\forall x \exists y (x = \frac{1}{y})$

counterexample: $x=2$; $2 = \frac{1}{y} \Rightarrow y = \frac{1}{2}$, which is not an integer

b) $\forall x \exists y (y^2 - x < 100)$

counterexample: $x = -101$; $y^2 - (-101) < 100 \Rightarrow y^2 + 101 < 100 \Rightarrow y^2 < -1$
which isn't possible because y^2 must be positive.

c) $\forall x \forall y (x^2 \neq y^3)$

counterexample: $x=0, y=0$; $x^2 = 0^2 = 0 = 0^3 = y^3$

(1.6)
1 If Socrates is human, then Socrates is mortal

Socrates is human

$\therefore$ Socrates is mortal

Modus Ponens, this is valid. The premises are true, so the conclusion is true

(1.6)
4 a) simplification

b) disjunctive syllogism

c) modus ponens

d) addition

e) hypothetical syllogism

1.6
5

w: "Randy works hard"

d: "Randy is a dull boy"

j: "Randy will get the job"

1. w Premise
2. $w \rightarrow d$ Premise
3. $d \rightarrow \neg j$ Premise
4. $w \rightarrow \neg j$ 2, 3 Hypothetical Syllogism
5. $\neg j$ 1, 4 Modus Ponens

1.6
9e

$P(x)$: "x is good for corporations"

$Q(x)$: "x is good for the United States"

$R(x)$: "x is good for you"

1. $\forall x P(x) \rightarrow Q(x)$ Premise
 2. $\forall x Q(x) \rightarrow R(x)$ Premise
 3. $P(\text{buying lots of stuff})$ Premise
 4. $P(\text{buying lots of stuff}) \rightarrow Q(\text{buying lots of stuff})$ 1, Universal Instantiation
 5. $Q(\text{buying lots of stuff}) \rightarrow R(\text{buying lots of stuff})$ 2, Universal Instantiation
 6. $Q(\text{buying lots of stuff})$ 3, 4 Modus Ponens
 7. $R(\text{buying lots of stuff})$ 5, 6 Modus Ponens
 8. $Q(\text{buying lots of stuff}) \wedge R(\text{buying lots of stuff})$ 7, 8 Conjunction
- Buying lots of stuff is good for the United States and buying lots of stuff is good for you.

(1.6)
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$p_1 \wedge p_2 \wedge \dots \wedge p_n$

q

$\therefore r$

is a valid argument

$p_1 \wedge p_2 \wedge \dots \wedge p_n$

$\therefore q \rightarrow r$

we want to show this is valid.

Proof: Suppose $p_1 \wedge p_2 \wedge \dots \wedge p_n$ is true, and suppose the first argument is valid.
We wish to show that $q \rightarrow r$ is true.
We know that q is either true or false.

Case 1: Suppose q is false. Then $q \rightarrow r$ is true, vacuously.

Case 2: Suppose q is true. Then by the first argument, we can conclude that r is true.

Thus, $q \rightarrow r$ is true since both q and r are true.

Since $q \rightarrow r$ is true in both cases, we can conclude that the argument form with premises $p_1 \wedge p_2 \wedge \dots \wedge p_n$, and conclusion $q \rightarrow r$ is valid.

(1.6)
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- a) Yes, universal instantiation, modus tollens
- b) No, universal instantiation, denying the hypothesis
- c) No, universal instantiation, affirming the conclusion
- d) Yes, universal instantiation, modus ponens

(1.6)
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Step 5 is incorrect. $Q(c)$ is said to be true for the same c that $P(c)$ is true for, which is not necessarily the case.
A different variable should be used to show that these could be true for different values.

(1.6 24) Steps 3 and 5 are incorrect. Simplification only applies to $\wedge$ statements, not $\vee$ statements.

- (1.6 26)
1. $\forall x (P(x) \rightarrow Q(x))$ Premise
 2. $\forall x (Q(x) \rightarrow R(x))$ Premise
 3. $R(a) \rightarrow Q(a)$ 1, Universal Instantiation
 4. $Q(a) \rightarrow R(a)$ 2, Universal Instantiation
 5. $P(a) \rightarrow R(a)$ 3, 4 Hypothetical Syllogism
 6. $\forall x (P(x) \rightarrow R(x))$ 5, Universal Generalization

- (1.6 35)
- a: Superman is able to prevent evil
w: Superman is willing to prevent evil
p: Superman prevents evil
i: Superman is impotent
m: Superman is malevolent
e: Superman exists

- | | | | |
|-----------------------------------|-------------------|------------------|---------------------|
| 1. $(a \wedge w) \rightarrow p$ | Premise | 11. a | 2, 9 Modus Tollens |
| 2. $\neg a \rightarrow i$ | Premise | 12. w | 3, 10 Modus Tollens |
| 3. $\neg w \rightarrow m$ | Premise | 13. $a \wedge w$ | 11, 12 Conjunction |
| 4. $\neg p$ | Premise | 14. p | 1, 13 Modus Ponens |
| 5. $e \rightarrow \neg(i \vee m)$ | Premise | | |
| 6. For contradiction, suppose e. | | | |
| 7. $\neg(i \vee m)$ | 5, 6 Modus Ponens | | |
| 8. $\neg i \wedge \neg m$ | 7 De Morgan's Law | | |
| 9. $\neg i$ | 8 Simplification | | |
| 10. $\neg m$ | 8 Simplification | | |

This is a contradiction: p and $\neg p$ can't both be true at the same time.

Thus, our assumption that e is true must be false.

Thus, Superman does not exist.

(1.7)
1) Proof (direct): Suppose x and y are odd integers.

Then $x = 2m + 1$ for some integer m , and

$y = 2n + 1$ for some integer n .

Thus, $x + y = 2m + 1 + 2n + 1 = 2m + 2n + 2 = 2(m + n + 1)$.

Since $m + n + 1$ is an integer, $x + y = 2(m + n + 1)$ is an even integer.

(1.7)
5) Proof (direct): Suppose $m+n$ and $n+p$ are even integers.

Then $m+n = 2a$ for some integer a , and

$n+p = 2b$ for some integer b .

We wish to show that $m+p$ is even.

Consider $(m+n) + (n+p) = 2a + 2b$

$$m + 2n + p = 2a + 2b$$

$$m + p = 2a + 2b - 2n$$

$$m + p = 2(a + b - n).$$

Since $a + b - n$ is an integer, $m + p = 2(a + b - n)$ is an even integer.

(1.7)
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From the hint, let's consider $(k+1)^2 - k^2$

$$= k^2 + 2k + 1 - k^2$$

$$= 2k + 1$$

Since k is any arbitrary integer, $2k+1$ is any arbitrary odd integer.
 So for any arbitrary odd integer $2k+1$, $2k+1$ is the difference of two squares, namely $k+1$ and k .

Proof (direct): Suppose x is an odd integer.

Then $x = 2k+1$, where k is some integer.

Then $x = 2k+1 = 2k+1 + k^2 - k^2$

$$= (k^2 + 2k + 1) - k^2$$

$$= (k+1)^2 - k^2$$

So x is the difference of two squares.

(1.7)
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Proof (contradiction): Suppose x is a rational number and suppose y is an irrational number.

For contradiction, suppose $x+y$ is rational.

Since x is rational, $x = \frac{a}{b}$, where a and b are integers, $b \neq 0$.

Since $x+y$ is rational, $x+y = \frac{c}{d}$, where c and d are integers, $d \neq 0$.

Then $y = (x+y) - x = \frac{c}{d} - \frac{a}{b} = \frac{cb - ad}{bd}$

Since $cb - ad$ and bd are integers, and since $b \neq 0$ and $d \neq 0$ makes $bd \neq 0$, $y = \frac{cb - ad}{bd}$ is a rational number.

This contradicts the fact that we assumed y is irrational.

Thus, $x+y$ is irrational.

(1.7)
15 Proof (contradiction): Suppose x is irrational

For contradiction, suppose $\sqrt{x}$ is rational.

Since $\sqrt{x}$ is rational, $\sqrt{x} = \frac{a}{b}$, where a and b are integers, and $b \neq 0$.

$$\text{Then } x = (\sqrt{x})^2 = \left(\frac{a}{b}\right)^2 = \frac{a^2}{b^2}$$

a^2 is an integer since a is an integer, and

b^2 is a nonzero integer since b is a nonzero integer.

Thus, x is rational.

But this contradicts our assumption that $\sqrt{x}$ is irrational.

Thus, $\sqrt{x}$ must be irrational.

